

guess based on the historical relationship between withdrawal rates, market valuations, and interest rates. It could end up being more or less (we won't know which for another thirty years). This estimated withdrawal rate has a 50 percent chance for success. To be conservative, a lower withdrawal rate is required to account for the additional random fluctuations from outside the model. This analysis further confirms the idea that the 4 percent withdrawal rate cannot be treated as safe for retirees in today's market environment. A 4 percent withdrawal rate should never be treated as an almost guaranteed annuitization rate from one's assets, and this is especially true for today's retirees.

These exercises about looking at the impacts of interest rates and market valuations illustrate that assumptions about future returns matter a great deal. This reemphasizes what we could already learn with the PMT formula. As has been the theme thus far, arbitrarily basing Monte Carlo simulations on historical averages may lead to overly optimistic results. Unfortunately, we do not know what the future will bring, and, ultimately, retirees should remain cautious and flexible.